

Sample of GitHub vs Notion API Comparison with evidence

1.1 Purpose of This Document

This document compares the structure and design of the GitHub and Notion APIs. It focuses on how each API represents data, relationships, content and user-defined structures, and how those design choices affect the developer experience.

The comparison is not intended to identify a single “better” API. Instead, it examines how each API is designed around the type of system it represents and where each approach is useful.

4.1.2 Structured Data Represented by Objects and Arrays

GitHub also uses JSON objects and arrays to represent structured data and collections.

Objects group related properties together. For example, an issue may contain a nested `user` object representing the user associated with that issue.

Arrays represent collections of values or objects. Properties such as `labels` and `assignees` can contain multiple items.

This creates an important distinction between different JSON value types:

- A string can represent a simple value such as `body`
- An object can group related properties
- An array can represent a collection of values or objects
- Numbers, Boolean values and `null` can represent other data states

My Test

Terminal command:

```
curl https://api.github.com/repos/octocat/Hello-World | jq
```

Response:

```
{  
  
  "id": 1296269,  
  
  "node_id": "MDEwOlJlcG9zaXRvcnkxMjk2MjY5",  
  
  "name": "Hello-World",
```

```
"full_name": "octocat/Hello-World",
"private": false,
"owner": {
  "login": "octocat",
  "id": 583231,
  "node_id": "MDQ6VXNlcjU4MzIzMQ==",
  "avatar_url": "https://avatars.githubusercontent.com/u/583231?v=4",
  "gravatar_id": "",
  "url": "https://api.github.com/users/octocat",
  "html_url": "https://github.com/octocat",
  "followers_url": "https://api.github.com/users/octocat/followers",
  "following_url": "https://api.github.com/users/octocat/following{/other_user}",
  "gists_url": "https://api.github.com/users/octocat/gists{/gist_id}",
  "starred_url": "https://api.github.com/users/octocat/starred{/owner}/{/repo}",
  "subscriptions_url": "https://api.github.com/users/octocat/subscriptions",
  "organizations_url": "https://api.github.com/users/octocat/orgs",
  "repos_url": "https://api.github.com/users/octocat/repos",
  "events_url": "https://api.github.com/users/octocat/events{/privacy}",
  "received_events_url": "https://api.github.com/users/octocat/received_events",
  "type": "User",
  "user_view_type": "public",
  "site_admin": false
},
"html_url": "https://github.com/octocat/Hello-World",
```

"description": "My first repository on GitHub!",

"fork": false,

"url": "https://api.github.com/repos/octocat/Hello-World",

"forks_url": "https://api.github.com/repos/octocat/Hello-World/forks",

"keys_url": "https://api.github.com/repos/octocat/Hello-World/keys{/key_id}",

"collaborators_url":
"https://api.github.com/repos/octocat/Hello-World/collaborators{/collaborator}",

"teams_url": "https://api.github.com/repos/octocat/Hello-World/teams",

"hooks_url": "https://api.github.com/repos/octocat/Hello-World/hooks",

"issue_events_url": "https://api.github.com/repos/octocat/Hello-World/issues/events{/number}",

"events_url": "https://api.github.com/repos/octocat/Hello-World/events",

"assignees_url": "https://api.github.com/repos/octocat/Hello-World/assignees{/user}",

"branches_url": "https://api.github.com/repos/octocat/Hello-World/branches{/branch}",

"tags_url": "https://api.github.com/repos/octocat/Hello-World/tags",

"blobs_url": "https://api.github.com/repos/octocat/Hello-World/git/blobs{/sha}",

"git_tags_url": "https://api.github.com/repos/octocat/Hello-World/git/tags{/sha}",

"git_refs_url": "https://api.github.com/repos/octocat/Hello-World/git/refs{/sha}",

"trees_url": "https://api.github.com/repos/octocat/Hello-World/git/trees{/sha}",

"statuses_url": "https://api.github.com/repos/octocat/Hello-World/statuses/{sha}",

"languages_url": "https://api.github.com/repos/octocat/Hello-World/languages",

"stargazers_url": "https://api.github.com/repos/octocat/Hello-World/stargazers",

"contributors_url": "https://api.github.com/repos/octocat/Hello-World/contributors",

"subscribers_url": "https://api.github.com/repos/octocat/Hello-World/subscribers",

"subscription_url": "https://api.github.com/repos/octocat/Hello-World/subscription",

"commits_url": "https://api.github.com/repos/octocat/Hello-World/commits{/sha}",

"git_commits_url": "https://api.github.com/repos/octocat/Hello-World/git/commits{/sha}",
"comments_url": "https://api.github.com/repos/octocat/Hello-World/comments{/number}",
"issue_comment_url":
"https://api.github.com/repos/octocat/Hello-World/issues/comments{/number}",
"contents_url": "https://api.github.com/repos/octocat/Hello-World/contents/{+path}",
"compare_url": "https://api.github.com/repos/octocat/Hello-World/compare/{base}...{head}",
"merges_url": "https://api.github.com/repos/octocat/Hello-World/merges",
"archive_url": "https://api.github.com/repos/octocat/Hello-World/{archive_format}/{ref}",
"downloads_url": "https://api.github.com/repos/octocat/Hello-World/downloads",
"issues_url": "https://api.github.com/repos/octocat/Hello-World/issues{/number}",
"pulls_url": "https://api.github.com/repos/octocat/Hello-World/pulls{/number}",
"milestones_url": "https://api.github.com/repos/octocat/Hello-World/milestones{/number}",
"notifications_url":
"https://api.github.com/repos/octocat/Hello-World/notifications{?since,all,participating}",
"labels_url": "https://api.github.com/repos/octocat/Hello-World/labels{/name}",
"releases_url": "https://api.github.com/repos/octocat/Hello-World/releases{/id}",
"deployments_url": "https://api.github.com/repos/octocat/Hello-World/deployments",
"created_at": "2011-01-26T19:01:12Z",
"updated_at": "2026-06-16T04:42:45Z",
"pushed_at": "2024-08-20T23:54:42Z",
"git_url": "git://github.com/octocat/Hello-World.git",
"ssh_url": "git@github.com:octocat/Hello-World.git",
"clone_url": "https://github.com/octocat/Hello-World.git",
"svn_url": "https://github.com/octocat/Hello-World",
"homepage": "",

"size": 1,
"stargazers_count": 3632,
"watchers_count": 3632,
"language": null,
"has_issues": true,
"has_projects": true,
"has_downloads": true,
"has_wiki": true,
"has_pages": false,
"has_discussions": false,
"forks_count": 5923,
"mirror_url": null,
"archived": false,
"disabled": false,
"open_issues_count": 6118,
"license": null,
"allow_forking": true,
"is_template": false,
"web_commit_signoff_required": false,
"has_pull_requests": true,
"pull_request_creation_policy": "all",
"topics": [],
"visibility": "public",
"forks": 5923,

```
"open_issues": 6118,  
"watchers": 3632,  
"default_branch": "master",  
"temp_clone_token": null,  
"network_count": 5923,  
"subscribers_count": 1723  
}
```

Evidence

- **id**: Internal repo identifier.
- **node_id**: Another identifier.
- **owner**: Direct support for nested owner user objects. This supports selective nesting for related resources in 4.1.3. This is the most obvious nested object. Because GitHub views the owner as user inside repo. Relationship between two resources.
- **login**: Human-readable username.
- **id**: User identifier.
- **avatar_url**: Profile image endpoint.
- **repos_url**: Inside owner object you already find links to Repos, Followers, Organizations, Events, Starred repos. This reinforces Relationship-oriented resource links.
- **html_url**: Human webpage. This exposes Human URL and API URL are separate.
- **description**: Supports repo metadata.
- **fork**: You don't mention this. Yet it's important repo metadata.
- **url**: API representation of this repo.
- **collaborators_url**: Collaborators is a URL.
- **assignees_url**: Array / collection.
- **issues_url**: Proves relationship links.
- **created_at** / **updated_at**: Supports timestamps.
- **pushed_at**: Omission from your section. This tracks code activity.
- **git_url**, **ssh_url**, **clone_url**, **svn_url**: All these **_url** fields are strong support for Relationship-oriented resource links and Multiple endpoint URL references.
- **stargazers_count**: Supports social metrics.
- **has_issues** through **has_discussions**: These describe repo capabilities.
- **forks_count**: Supports social metrics.

- `topics`: Proves arrays are used for collections. Even though the collection is empty. GitHub still returns `[]` because multiple topics are possible.
 - `default_branch`: Core repo metadata.
 - `temp_clone_token`: `null`.
 - The response also contains embedded objects, URL references and collections. This relates to supporting 4.1.3. Also `user`, `repo_url` and `comments_url` supports.
 - Apparently labels and assignees are array / collections. Reactions is embedded object. This relates to supporting 4.1.3.
-

4.2.2 Pagination

Large collections are commonly divided across multiple responses rather than returned in one very large response. GitHub uses pagination to allow clients to request additional pages of results.

Pagination information can be provided through the HTTP response headers.

Example response header:

Link: <https://api.github.com/user/repos?page=2>; rel="next"

The `rel="next"` relationship indicates that the supplied URL can be used to retrieve the next page.

My Test

Response headers:

Link: <<https://api.github.com/user/repos?page=2>>; rel="next"

HTTP/2 200

date: Thu, 18 Jun 2026 06:35:15 GMT

content-type: application/json; charset=utf-8

cache-control: public, max-age=60, s-maxage=60

vary: Accept,Accept-Encoding, Accept, X-Requested-With

etag: W/"9b56442ae87373a9f6da29b7e37fe00abb08b16961e9a91cf340d8da5905e4d7"

x-github-media-type: github.v3; format=json

link: <https://api.github.com/user/583231/repos?per_page=2&page=2>; rel="next",
<https://api.github.com/user/583231/repos?per_page=2&page=4>; rel="last"

x-github-api-version-selected: 2022-11-28

access-control-expose-headers: ETag, Link, Location, Retry-After, X-GitHub-OTP,
X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Used, X-RateLimit-Resource,
X-RateLimit-Reset, X-OAuth-Scopes, X-Accepted-OAuth-Scopes, X-Poll-Interval,
X-GitHub-Media-Type, X-GitHub-SSO, X-GitHub-Request-Id, Deprecation, Sunset, Warning

access-control-allow-origin: *

strict-transport-security: max-age=31536000; includeSubdomains; preload

x-frame-options: deny

x-content-type-options: nosniff

x-xss-protection: 0

referrer-policy: origin-when-cross-origin, strict-origin-when-cross-origin

content-security-policy: default-src 'none'

server: github.com

accept-ranges: bytes

x-ratelimit-limit: 60

x-ratelimit-remaining: 59

x-ratelimit-used: 1

x-ratelimit-resource: core

x-ratelimit-reset: 1781768115

content-length: 11976

x-github-request-id: C097:214FD8:A8BEB5:BB6055:6A3391A3

JSON response:

```
[
  {
    "id": 132935648,
    "node_id": "MDEwOlJlcG9zaXRvcnkxMzI5MzU2NDg=",
    "name": "boysenberry-repo-1",
    "full_name": "octocat/boysenberry-repo-1",
    "private": false,
    "owner": {
      "login": "octocat",
      "id": 583231,
      "node_id": "MDQ6VXNlcjU4MzIzMQ==",
      "avatar_url": "https://avatars.githubusercontent.com/u/583231?v=4",
      "gravatar_id": "",
      "url": "https://api.github.com/users/octocat",
      "html_url": "https://github.com/octocat",
      "followers_url": "https://api.github.com/users/octocat/followers",
      "following_url": "https://api.github.com/users/octocat/following{/other_user}",
      "gists_url": "https://api.github.com/users/octocat/gists{/gist_id}",
      "starred_url": "https://api.github.com/users/octocat/starred{/owner}/{/repo}",
      "subscriptions_url": "https://api.github.com/users/octocat/subscriptions",
      "organizations_url": "https://api.github.com/users/octocat/orgs",
      "repos_url": "https://api.github.com/users/octocat/repos",
      "events_url": "https://api.github.com/users/octocat/events{/privacy}",
```

```
"received_events_url": "https://api.github.com/users/octocat/received_events",
"type": "User",
"user_view_type": "public",
"site_admin": false
},
"html_url": "https://github.com/octocat/boysenberry-repo-1",
"description": "Testing",
"fork": true,
"url": "https://api.github.com/repos/octocat/boysenberry-repo-1",
"forks_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/forks",
"keys_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/keys{/key_id}",
"collaborators_url":
"https://api.github.com/repos/octocat/boysenberry-repo-1/collaborators{/collaborator}",
"teams_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/teams",
"hooks_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/hooks",
"issue_events_url":
"https://api.github.com/repos/octocat/boysenberry-repo-1/issues/events{/number}",
"events_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/events",
"assignees_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/assignees{/user}",
"branches_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/branches{/branch}",
"tags_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/tags",
"blobs_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/git/blobs{/sha}",
"git_tags_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/git/tags{/sha}",
"git_refs_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/git/refs{/sha}",
"trees_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/git/trees{/sha}",
```

"statuses_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/statuses/{sha}",
"languages_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/languages",
"stargazers_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/stargazers",
"contributors_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/contributors",
"subscribers_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/subscribers",
"subscription_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/subscription",
"commits_url": "https://api.github.com/repos/octocat/boysenberry-repo-1/commits{/sha}",
"git_commits_url":
"https://api.github.com/repos/octocat/boysenberry-repo-1/git/commits{/sha}",
"comments_url":
"https://api.github.com/repos/octocat/boysenberry-repo-1/comments{/number}",
"issue_comme_